Lecture 14: Single Phase AC Circuits

A bulk of electric power generated in power plant, throughout the world is in form of ac (alternating current) or alternating voltage quantities. Alternating quantities are sinusoidal voltage and current variations, specified by magnitude and frequency.

Why Sinusoidally varying voltage or current quantities?

There are many technological and economical reasons such as

- A sinusoidal signal is simplest periodic signal in nature eg, pendulum motion, tuning fork, LC-oscillations, generator voltage output etc.
- A sinusoid function is a smooth function ⇒ Its derivative, integral retain the same nature.
- Electric generators in power system generate sinusoidal electrical voltage, due to circular movement of rotor and arrangement of coils.

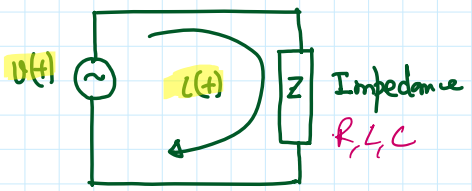
Hence, study of AC circuits is essential to understand the electrical systems. Although at power level 3ϕ circuits are normally used. However to start with AC network, first we will discuss single phase followed by 3ϕ systems.

Sinusoidal Functions - Terminology:

Consider a circuit, excited by a sinusoidal voltage source.

$$v(t) = V_m \sin \omega t$$

$$i(t) = I_m \sin(\omega t - \phi)$$



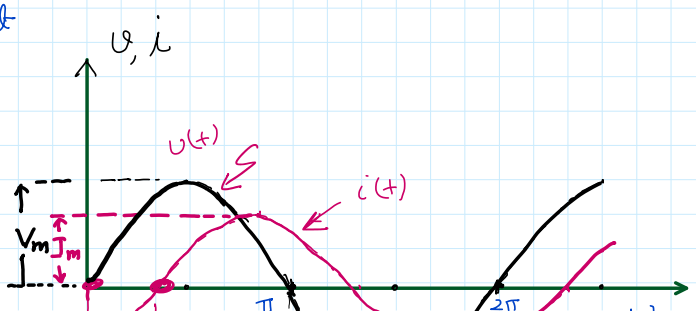
V_m = Maximum or peak value of voltage

I_m = Maximum or peak value of current

ω = Radian frequency of voltage and current waveform

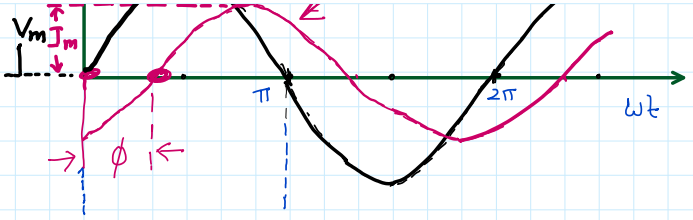
$= 2\pi f$

$f = \frac{\omega}{2\pi}$, $T = \frac{2\pi}{\omega}$ Sec.



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ϕ = Phase shift of current with respect to $v(t)$.



Average and RMS Quantities

For sinusoidal voltage and current quantity, the average value over a cycle is zero. This does not add any meaningful description. Hence we take half cycle average, as follows.

$$V_{avg\,1/2} = \frac{1}{T/2}\int_0^{T/2} v(t)\,dt = \frac{1}{\pi}\int_0^{\pi} (V_m \sin\omega t)\, d(\omega t)$$

$$= \frac{V_m}{\pi}\left|-\cos\omega t\right|_0^{\pi} = \frac{2V_m}{\pi}$$

Similarly, $I_{avg\,1/2} = \frac{1}{T/2}\int_0^{T/2} i(t)\,dt = \frac{2I_m}{\pi}$

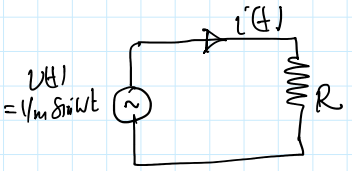
Thus, we have

$$V_{avg\,\frac{1}{2}} = \frac{2V_m}{\pi}$$

$$I_{avg\,\frac{1}{2}} = \frac{2I_m}{\pi}$$

Root Mean Square (RMS) Quantity:

Basically Root Mean Square quantity of ac voltage or current represents its dc equivalents.



$p(t) = ?$

$i(t) = \frac{v(t)}{R} = \frac{V_m \sin\omega t}{R}$

$i(t) = \frac{V_m}{R}\sin\omega t$

$i(t) = I_m \sin\omega t$

$p(t) = V_m \sin\omega t * \frac{V_m \sin\omega t}{R} = \frac{(V_m \sin\omega t)^2}{R}$

$= i^2 * R = \frac{v^2}{R}$

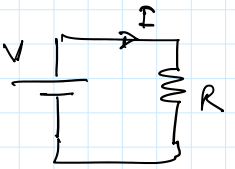
$p(t) = (I_m \sin\omega t)^2 * R$

$I_m = (V_m / R)$

P = Real power dissipated in R

$$P = \frac{1}{T}\int_0^T (I_m \sin \omega t)^2 * R\, dt$$

$$= I_{eff}^2 * R$$



$$I_{eff} = I_{rms} = I = \left[\frac{1}{T}\int_0^T i^2(t)\, dt\right]^{1/2}$$

$$V_{eff} = V_{rms} = V = \left[\frac{1}{T}\int_0^T v^2(t)\, dt\right]^{1/2}$$

$$= \left[\frac{1}{T} \cdot \int_0^T V_m^2 \sin^2 \omega t\, dt\right]^{1/2}$$

$$= \left[\frac{1}{T} * \int_0^T V_m^2 \frac{(1 - \cos 2\omega t)}{2}\, dt\right]^{1/2}$$

$$= \left(\frac{1}{T} * \frac{V_m^2}{2} \times T\right)^{1/2} = \frac{V_m}{\sqrt{2}}$$

Thus, $V_{eff} = V_{rms} = V = \frac{V_m}{\sqrt{2}}$

and $I_{eff} = I_{rms} = I = \frac{I_m}{\sqrt{2}}$

This can also be understood in terms of real power (P).

A sinusoidal voltage $v(t)$ supplying a resistive load, the instantaneous power, $p(t)$ is given by

$$p(t) = v(t) \cdot i(t) = i(t)\, R\, i(t) = i^2 R$$

Average power consumed by resistance R is given by

$$P = \frac{1}{T}\int_0^T p(t)\, dt = \frac{1}{T}\int_0^T i^2 R\, dt = \frac{R}{2\pi}\int_0^{2\pi} I_m^2 \sin^2(\omega t - \phi)\, d(\omega t)$$

$$= \frac{R}{2\pi} * \int_0^{2\pi} \frac{I_m^2 \{1 - \cos k(\omega t - \phi)\}}{2}\, dt$$

$$= \frac{1}{2\pi} \cdot \frac{I_m^2}{2} \cdot 2\pi = \frac{I_m^2}{2} R$$

$$= \left(\frac{I}{\sqrt{2}}\right)^2 * R = I^2 R$$

Thus $P = I^2R$ for I as Rms value. The expression is similar to dc network, where $P_{dc} = I_{dc}^2 * R$.

Power Quantities in AC systems:

Using Rms values, we express voltage and current quantity,

$$v(t) = \sqrt{2}\, V \sin \omega t$$
$$i(t) = \sqrt{2}\, I \sin(\omega t - \phi)$$

Instantaneous power, $p(t)$ is defined as,

$$p(t) = v(t)\, i(t) = V_m \sin \omega t \cdot I_m \sin(\omega t - \phi)$$

$$= \frac{V_m I_m}{2} \left[2 \sin \omega t \cdot \sin(\omega t - \phi) \right]$$

$$= \frac{V_m I_m}{2} \left[\cos \phi - \cos(2\omega t - \phi) \right]$$

$$= \frac{V_m I_m}{2} \cos \phi - \frac{V_m I_m}{2} \cos(2\omega t - \phi)$$

$$= VI \cos \phi - VI \cos(2\omega t - \phi)$$

$$p(t) = \bar{p} + \tilde{p}$$

(Real power) + oscillating power

Another level classification is achieved through expanding of term $\cos(2\omega t - \phi)$, as follows.

$$p(t) = VI \cos\phi - (VI \cos 2\omega t \cos\phi + \sin 2\omega t \sin\phi)$$

$$= VI \cos\phi (1 - \cos 2\omega t) - VI \sin\phi \sin 2\omega t$$

$$\doteq p_{active} + p_{reactive}$$

$$= p_{unidirectional} + p_{reactive}$$

Real Power $= P = \frac{1}{T} \int_0^T p(t)\, dt = \frac{1}{T} \int_0^T p_{active}(t)\, dt = VI \cos\phi$

Reactive Power $= Q \triangleq$ maximum value of $p_{reactive}$

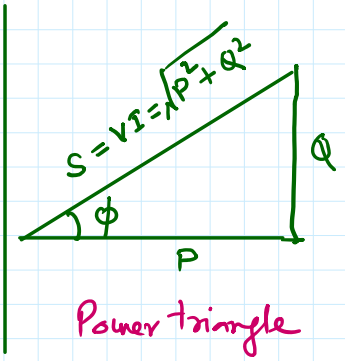
Apparent Power $= S =$ Product of Rms value of voltage and Rms value of current

$$= V_{rms} I_{rms} = VI$$

$= V_{rms} I_{rms} = VI$

Thus $S = VI = \sqrt{(VI)^2 \cos^2\phi + (VI)^2 \sin^2\phi} = \sqrt{P^2 + Q^2}$

This relation is indicated by power triangle as shown below.



Power triangle

Now we define term power factor or utilization factor, as follows.

$$pf = \frac{\text{Real Power}}{\text{Apparent Power}} = \frac{P}{S}$$

$$= \frac{VI\cos\phi}{VI} = \cos\phi$$

$$\boxed{pf = P/S = \cos\phi}$$